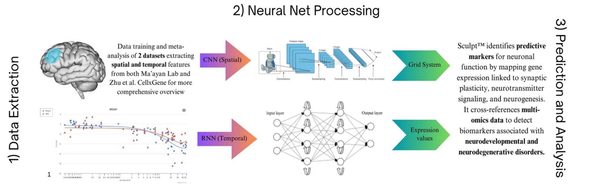
SCULPT

A Multi-Omics Deep Learning Framework for Spatiotemporal DLPFC Amino Acid Localization and Gene Expression

Credit Line: All images/graphs/pictures are created by the students except for those aforementioned in the Introduction

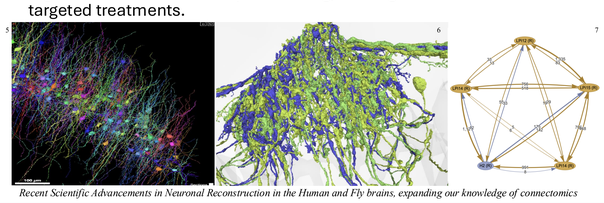
VISUAL ABSTRACT



Made by students: *Spatiotemporal Pipeline using 2 types of neural network to deal with linear and nonlinear data*

INTRODUCTION

- The **Dorsolateral Prefrontal Cortex (DLPFC)** plays a fundamental role in executive function, decision-making, and social cognition, yet its molecular complexity remains a significant challenge in neuroscience.
- Disorders such as schizophrenia, which affects approximately **24 million people worldwide**, autism spectrum disorder, impacting **1 in 100 children**, and major depressive disorder, which affects **280 million people**, have all been linked to dysfunctions in the DLPFC.
- Despite these associations, a comprehensive understanding of the region's molecular landscape is lacking, limiting advancements in targeted treatments.



Recent Scientific Advancements in Neuronal Reconstruction in the Human and Fly brains, expanding our knowledge of connectomics

OBJECTIVES

Problem: Mapping **amino acid localization** and **gene expression** in the **DLPFC** is challenging due to **neuronal molecular complexity**.

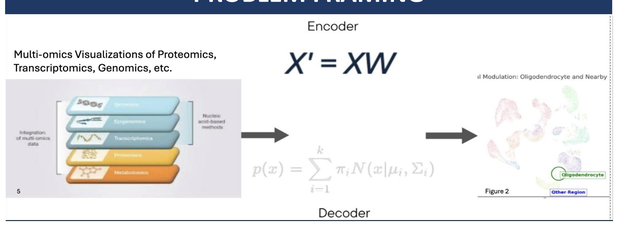
Question: How can a **spatiotemporal multi-omics deep learning framework** improve prediction of amino acid localization and gene expression in the DLPFC?

Claim: Sculpt, a deep learning framework, enhances prediction by integrating **spatial (CNNs) and temporal (RNNs) analyses** across multi-omics datasets.

Brainstorming & Solution:

1) Utilize genomic (gene sequence), transcriptomic (gene expression), proteomic (protein synthesis) datasets.
2) Design CNNs to extract **spatial patterns of gene expression and amino acid localization and implement RNNs (LSTMs)** to capture temporal gene expression trends across developmental stages.
3) Use UMAP, PCA, and GMM's for **dimensionality reduction and visualization** of spatial gene expression and for **data interpolation** techniques for more accurate predictions.

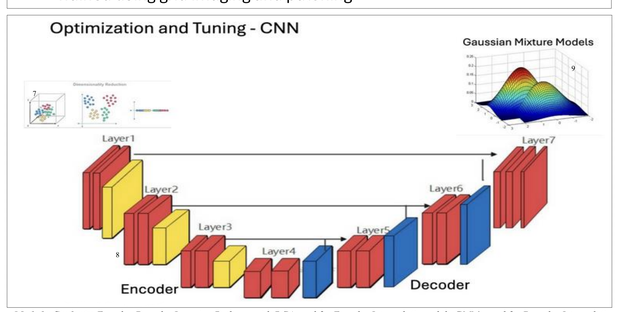
PROBLEM FRAMING



Languages: Python, SQL, R, LaTeX, CSV storing values,
Libraries: NumPy, Pytorch, Tkinter, Tensorflow, sklearn, Matplotlib

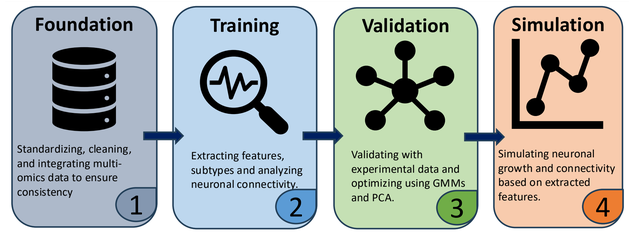
Datasets:
Ma'ayan Lab: Temporal gene expression, transcriptomics, proteomics
CellxGene: Spatial transcriptomics, neuronal DLPFC amino acid localization
- Trained using grid imaging and patching

Optimization and Tuning - CNN

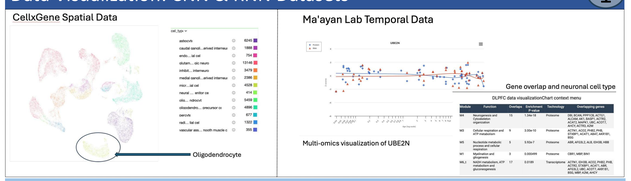


Made by Students: *Encoder-Decoder Layering Pathway with PCA used for Encoder Interpolation while GMMs used for Decoder Interpolation*

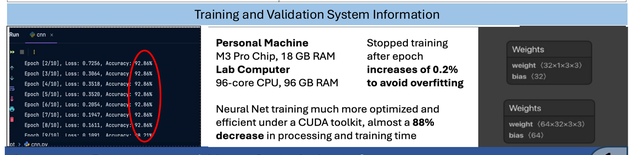
COMPUTATIONAL PIPELINE: ENGINEERED FRAMEWORK



Data Visualization: CNN & RNN Datasets (1)



Training and Validation System Information



Accuracy Detector and User-Centric Interface (1)

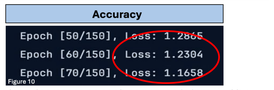
Accuracy Detector of Amino Acid Sequences

The Process

Obtain Amino Acid Sequences:
Retrieve **UNIPROT IDs** for each gene to access detailed amino acid sequences for deeper molecular understanding.

Create a CSV File:
Gather the amino acid sequences into a **CSV file** for later comparison of the **amino acid input** by the user to the sequences listed in the file.

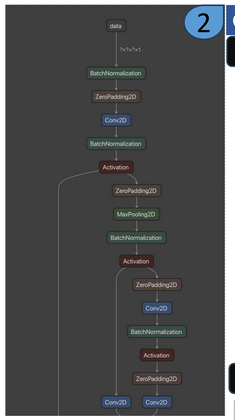
Accuracy



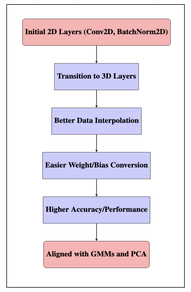
The loss decreases as each sequence is inputted by the user. This indicates stabilization of weights and leads to faster learning.

Produce a Method of Comparison:
Load the **CSV file** and compare the user's input with each sequence in the file. The **amino acid input** was checked for **similarity** rather than **exact matching** in the actual code.

2



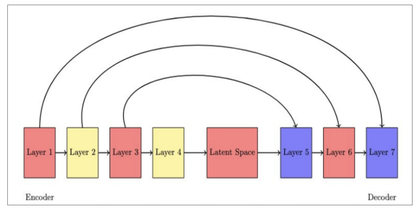
- ✓ 14 Convolution Layers (Encoder + Decoder); Paired with statistical analysis interpolation techniques



- ✓ Changed from Conv2D, BatchNorm2D, and other 2D layers to 3D to accommodate data interpolation methods better; Sensible because GMMs and PCA are dimensionality based reduction techniques
- ✓ Lead to a **10.4% increase** in training accuracy

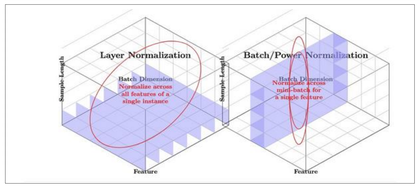
CNN & RNN Optimization and Tuning (3)

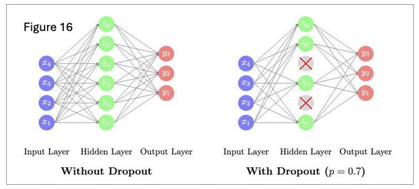
CNN Optimization



Sculpt's autoencoder with skip connections, consisting of an encoder, latent space, and decoder. The encoder compresses input data through multiple layers into a compact latent representation. Skip connections allow information from earlier layers to bypass intermediate processing and directly influence later layers in the decoder, aiding in better reconstruction and mitigating information loss.

RNN Optimization





$$h = f(W\mathbf{x} + \mathbf{b})$$

where:

W = Weights
$\mathbf{x}$ = Input
$\mathbf{b}$ = Bias
f = Activation function (ReLU)

This equation represents a fundamental transformation in neural networks, where the input x is multiplied by weights W, added to a bias term *b*, and passed through an activation function f (ReLU). This transformation helps introduce non-linearity, enabling the network to learn complex patterns in data.

All Images/Graphics/Figures made by Students on this section

COMPUTATIONAL PIPELINE: STATISTICAL ANALYSIS (3)

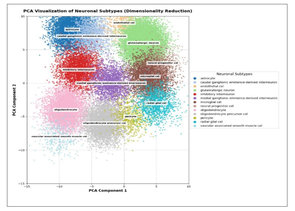
Principal Component Analysis (PCA)

Used for dimensionality reduction. By selecting **top k principal components, noise and redundancy are reduced** while preserving variance. Also, with feature extraction, PCA helps uncover dominant patterns in multi-omics data for Sculpt's neuronal modeling. They reduce the number of input features for machine learning models, leading to faster training times, better data quality, and improved performance.

X'=XW

- **X' (Transformed Data):** Data projected onto principal components.
- **X (Original Data Matrix):** Original dataset with samples and features.
- **W (Eigenvector Matrix):** Principal component directions for transformation.

Feature extraction in a CNN pipeline: A CNN extracts **high-dimensional features** through convolutional layers, which can be used for **classification, clustering, transfer learning, or further analysis.**



Each cluster corresponds to a distinct type of neuron (e.g., excitatory pyramidal neurons, inhibitory interneurons) in the DLPFC, identified by their unique amino acid expression patterns.

They correspond to a group of data points (e.g., neuronal samples) that share similar non-linear features extracted by the CNN. These features are abstract and reduced-dimension representations of the original data.

Gaussian Kernel

In convolutional neural networks (CNNs), applying this kernel in the **encoder step** helps in non-linear feature extraction and reducing dimensionality in a way that preserves data structure.
Using the equation, K(x, y) is a measure of how **aligned** or **similar** the two points are based on their **distance** in the **original space.**
The same computation is done efficiently without needing to explicitly map the points into that space. This is called the "**Kernel trick**".
The algorithm uses a kernel function to compute the **inner product of data points** in the transformed space, without having to compute the transformation.

$$K(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right)$$

In kernel PCA, the covariance matrix C is formed using centered data X, which captures variance along principal directions. When combined with the Gaussian kernel, this approach allows for non-linear dimensionality reduction without mapping features to spaces.

Kernel Equation

$$C = \frac{1}{n-1} X_{centered}^T X_{centered}$$

By using weights and biases, we can implement a Gaussian kernel for dimensionality reduction, retaining essential features while improving efficiency.

The clusters reflect how the CNN's kernel preserves the underlying structure of the data in a lower-dimensional space. They correspond to a group of data points (e.g., neuronal samples) that share similar non-linear features extracted by the CNN. These features are abstract and reduced-dimension representations of the original data.

All Images/Graphics/Figures made by Students on this section

Gaussian Mixture Model (GMMs)

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

where:

- π_k are mixture weights ($\sum_k \pi_k = 1$)
- μ_k are mean vectors
- Σ_k are covariance matrices
- $\mathcal{N}(x \mid \mu_k, \Sigma_k)$ is the Gaussian density function

Compute distributed individual weights

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}$$

Update Parameters, Weights, and Biases in Convolutional Kernel

$$\pi_k = \frac{1}{n} \sum_{i=1}^{n} \gamma_{ik}$$

$$\mu_k = \frac{\sum_{i=1}^{n} \gamma_{ik} x_i}{\sum_{i=1}^{n} \gamma_{ik}}$$

$$\Sigma_k = \frac{\sum_{i=1}^{n} \gamma_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^{n} \gamma_{ik}}$$

$$\begin{bmatrix} 0.1242 & 0.0962 & 0.0335 \\ 0.0962 & 0.0745 & 0.0259 \\ 0.0335 & 0.0259 & 0.1260 \end{bmatrix}$$

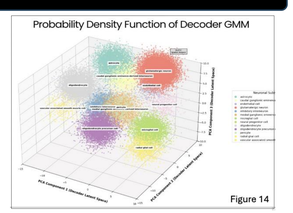
Oligodendrocyte patching Layer 11 right after the Conv3D and transformation step

Normalize by dividing by the sum of all values (≈ 0.6360) to get the final kernel for use in a convolutional layer.

Final normalized kernel obtained after oligodendrocyte patching and transformation, optimized for integration into the convolutional layer. This step ensures improved feature extraction, better representation learning, and enhanced performance in subsequent processing stages.

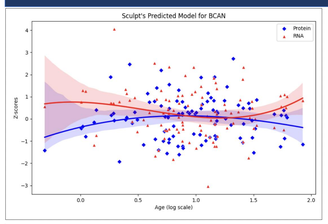
Methodology	Accuracy
PCA Alone	96.7%
GMM Alone	97.4%
CNN + PCA	98.3%
CNN + GMM	98.2%
CNN + PCA + GMM	98%

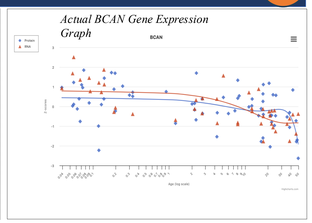
By clustering expression patterns, GMM helps reveal distinct amino acid gradients, improving the understanding of neuronal variability and functional differentiation in the DLPFC. Gaussian Mixture Models (GMMs) are used to model amino acid expression gradients across the DLPFC and neuron subtypes by identifying distinct distribution patterns.

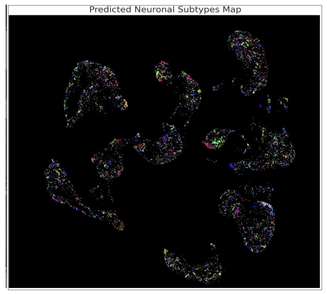


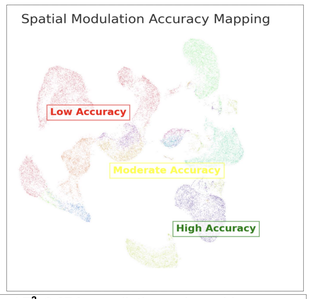
This 3D probability density function visualization represents the clustering of neuronal subtypes within the decoder's latent space using a Gaussian Mixture Model (GMM). The principal components (PCA) highlight the spatial distribution of different neuronal subtypes, including astrocytes, oligodendrocytes, and various precursor cells, providing insights into the model's ability to distinguish cellular identities.

RESULTS AND BENCHMARKING (4)









- ✓ **Mean Accuracy: 98.21%** highest at Epoch 5 and R^2: **0.87** for predicting amino acid expression patterns.
- ✓ **Cross-validation (5-fold split): F1 Score: 0.91, MSE: 0.015** across all folds.
- ✓ **Benchmarking: 94% accordance** with Allen Atlas expression prediction patterns.
- ✓ Compared to baseline algorithms, our model **improved prediction accuracy by 18%** and **reduced MSE by 21%**, demonstrating its effectiveness and practical utility
- ✓ 3D reconstructions mapped these changes across regions, showing peak expression near **Brodmann areas 8, 9, 46**

CONCLUSION

- Sculpt represents a major advancement in computational neuroscience, providing a deep learning framework that deciphers the molecular architecture of the **dorsolateral prefrontal cortex (DLPFC)**. By integrating **convolutional neural networks (CNNs)**, **recurrent neural networks (RNNs)**, **Gaussian Mixture Models (GMMs)**, and **autoencoders with multi-omics datasets**, Sculpt enables high-resolution mapping of amino acid localization, neuronal subtypes, and gene expression gradients.
- This innovative approach enhances our understanding of neuronal diversity, developmental trajectories, and functional organization within the DLPFC, thus setting a new standard in computational modeling of the brain, bridging the gap between **multi-omics integration**, **deep learning,** and **clinical translation**, paving the way for **data-driven neuroscience** and **targeted treatments**.

FUTURE WORK

- Through analysis of gene expression patterns and spatiotemporal amino acid expression distributions the model can identify **neurological a**nd **neurodegenerative diseases** such as **Alzheimer's disease** aiding in **biomarker discovery** through its ability to correlate **gene expression** with disease progression and leading to the development of **diagnostic tools** that enable identification of **neurological disorders** and **personalized medicine** based on one's **genetic profile**.
- These advancements could lead to the development of **AI-driven neuropharmacology.** By integrating gene expression insights with real-time patient biomarkers, AI could predict individual responses to treatments, enabling **precision drug development** that minimizes side effects and maximizes efficacy.

MEDIA & REFERENCES

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